Ram Moment Certificates for the Ising Random Perceptron

Nonlinear Gaussian Recursions, Variational Enclosures, and Disorder Extensions

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Abstract

We investigate the analytic mechanisms in OpenAI's manuscript on the free energy of the Ising random perceptron, separating its proposed thermodynamic formula from independently proved local and conditional consequences. Ram Calculus is relevant to the normalized nonlinear Gaussian integral, finite moment remainders, complete state derivatives, and propagation of numerical uncertainty. We give an amplitude-moment certificate that remains regular at zero hierarchy coefficient, an additive error budget across a finite hierarchy, and a finite-trial lower/upper enclosure of the variational value without silently truncating its unbounded entropy dual. We derive a quantitative third-moment disorder replacement estimate and its bounded almost-everywhere-continuous extension. These results retain the actual Ising prior, the terminal coefficient-one transform, and the distinct roles of Gaussian and discrete disorder. The source's overlap, synchronization, contact-point and cavity arguments are not replaced by moment coordinates, and finite diagnostics do not certify their infinite-volume conclusion. Contribution assessment: C3/5, the contribution is a notable source-specific certification and extension framework with explicit hypotheses, not a new unconditional proof of the full free-energy formula or a demonstrated computational speedup.

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1 Problem, Scope, and Evidence

The assigned source is *The free energy of the Ising random perceptron*, September 24, 2026, at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a` of the OpenAI mathematics release [1]. We keep its identifier M412; its spherical companion M415 is a different report. The applicability assessment is L4/5. Applicability measures the fit of a mathematical method, not the correctness of a source theorem or the size of a realized contribution.

The analysis has three logically different outputs. First, it reconstructs the source's normalization and the Gaussian recursion identities. Second, it proves finite and functional estimates whose proofs do not require the source's thermodynamic theorem. Third, it explains the conditional consequence for limiting pressure if that theorem and its probabilistic inputs hold. Classical Taylor, Gaussian integration, convexity, contraction, and replacement tools remain credited as classical; applying them in Ram moment coordinates is not a historical-priority claim.

The actual private passages used are the finite local-moment and measure-kernel statements in revised Volume 1, Chapter 3; the full state derivative and tail cautions in the nonlinear Urysohn report; and the composition and residual-ball statements in the nonlinear stability report [2, 3, 4]. These are narrower than an assertion that every theorem in the private corpus has been verified. Detailed passage and source records accompany this report. No private manuscript content is needed in a public literature query.

2 State Space and the Exact Variational Target

Let ν_N be uniform *probability* on $\{-1,1\}^N$, $M_N = \lfloor \alpha N \rfloor$, $\alpha > 0$, and let $g^a = (g_i^a)_{i=1}^N$ be independent standard Gaussian patterns. For a finite bounded Borel $f : \mathbb{R} \to \mathbb{R}$, define

$$b_a(x) = N^{-1/2} \sum_i g_i^a x_i, \qquad p_N(f) = \frac{1}{N} \log \int \exp\left(\sum_{a=1}^{M_N} f(b_a(x)) \right) \nu_N(dx). \tag{1}$$

Counting measure adds $\log 2$ to this pressure. Physical free energy with $f = \beta\phi$ is $-p_N/\beta$ under the corresponding convention. There is no spatial boundary or gauge condition: the finite state space is the cube with its stated measure. The scalar Gaussian operators below act on functions of a real cavity field, not on spatial PDE solutions.

An overlap path $q \in \mathcal{Q}$ is an almost-everywhere equivalence class of nondecreasing functions $(0,1) \to [0,1]$. A field path h is nonnegative and nondecreasing; the initial entropy definition takes all finite-valued step paths without a common upper cap. Lebesgue measure is used in every path integral. On a partition $0 = \zeta_0 < \cdots < \zeta_k = 1$, put $w_i = \zeta_{i+1} - \zeta_i$. For $s \geq 0$ and $0 \leq d \leq 1$, let

$$\mathcal{T}_{s,d} U(x) = \begin{cases} d^{-1} \log \mathbb{E} e^{dU(x + \sqrt{s}G)}, & d > 0, \\ \mathbb{E} U(x + \sqrt{s}G), & d = 0. \end{cases} \tag{2}$$

For bounded Borel U it is well defined, including at $s = 0$. For functions of at most linear growth it is well defined by Gaussian exponential integrability. These are two different admissible classes; an arbitrary L^2 function need not have the required exponential moment. Differentiation in x at zero variance additionally requires the relevant derivatives of U.

For the pattern recursion append $q_{-1} = 0, q_k = 1$, set $U_k = f$, and recursively put

$$U_{i-1} = \mathcal{T}_{q_i - q_{i-1}, \zeta_i} U_i \quad (i = k, \dots, 0), \qquad V_f(q) = U_{-1}(0). \tag{3}$$

For the field recursion append $h_{-1} = 0$, begin with $U_{k-1}(x) = \log\cosh x - h_{k-1}/2$, and use

$$U_{i-1} = \mathcal{T}_{h_i - h_{i-1}, \zeta_i} U_i \quad (i = k-1, \dots, 0), \qquad \ell(h) = U_{-1}(0). \tag{4}$$

The pattern recursion has a final coefficient-one Gaussian transform; the field recursion does not. Both have an initial coefficient-zero transform. Repeated path values insert identity transforms, not additional noise. Refining partitions in this way changes no value.

The source defines

$$S_{\mathrm{I}}(q) = \sup_{h \text{ step}} \left(\ell(h) + \frac{1}{2} \int_0^1 h(u) q(u)\, du \right), \qquad \mathcal{P}_f(\alpha) = \inf_{q \in \mathcal{Q}} \{ \alpha V_f(q) + S_{\mathrm{I}}(q) \}. \tag{5}$$

Its main theorem asserts $\mathbb{E} p_N(f) \to \mathcal{P}_f(\alpha)$ and convergence in probability for every bounded Borel f and every fixed $\alpha > 0$. The assertion is not the usual Gaussian mixed p-spin formula: $\sum_a f(b_a(x))$ is generally a non-Gaussian process indexed by spins. In particular, a Gaussian disorder matrix does not make a nonlinear Hamiltonian Gaussian.

The elementary bounds are $S_{\mathrm{I}} \geq 0$, $S_{\mathrm{I}}(0) = 0$, and $|V_f| \leq \|f\|_\infty$. Thus $-\alpha \|f\|_\infty \leq \mathcal{P}_f(\alpha) \leq \alpha \log \mathbb{E} e^{f(G)}$. The entropy can be extended valued. A finite list of q, h trials does not by itself evaluate the infimum and supremum in (5).

3 Normalized Gaussian Calculus and Its Ram Interpretation

3.1 The complete derivative

For $W = \mathcal{T}_{s,d} U$, define the probability kernel

$$K_{s,d,U}(x, dy) = e^{d(U(y) - W(x))} \mathcal{N}(x, s)(dy),$$

with the ordinary Gaussian measure at $d = 0$. For bounded perturbations A, B, differentiation of the normalized integral gives

$$D_U \mathcal{T}_{s,d}(U)[A] = KA, \tag{6}$$

$$D_U^2 \mathcal{T}_{s,d}(U)[A, B] = d\{K(AB) - (KA)(KB)\}. \tag{7}$$

Dominated differentiation follows on a bounded state ball. For linear growth terminals, bounded perturbations and local Gaussian exponential majorants suffice. The covariance term is the derivative of the normalizer as well as the numerator. Freezing that normalizer would give a different, incorrect nonlinear derivative.

The source/observation-state derivative rule of [3] enforces the same discipline. It does not eliminate the normalizing logarithm. Indeed $\|D_U \mathcal{T}\|_{\infty \to \infty} \leq 1$ and $D_U^2 \mathcal{T}[A, A] \geq 0$ for $d \geq 0$. These are derivatives in the terminal function; they do not assert convexity in covariance paths, whose increments enter several transforms with different signs.

For smooth U with integrable dominating derivatives, the spatial and variance identities are

$$W' = KU', \qquad W'' = KU'' + d\{K(U'^2) - (KU')^2\}, \tag{8}$$

$$\partial_s W = \tfrac{1}{2}\{W'' + d(W')^2\} = \tfrac{1}{2}K\{U'' + d(U')^2\}. \tag{9}$$

One proves the last equation by applying the heat equation to $e^{dW} = \mathbb{E}e^{dU(x+\sqrt{s}G)}$, then dividing by its strictly positive value. The $d = 0$ identity is the heat equation. At $s = 0$ it is a one-sided identity for a sufficiently smooth terminal function, not a Taylor theorem for arbitrary bounded Borel data.

3.2 Finite spatial moments, with a remainder

The Ram finite local theorem [2] applies to the positive Gaussian displacement measure as well as to a density. If $F = e^{dU}$ has a global bound B_{2r+2} on its $(2r+2)$nd derivative, then

$$\mathbb{E}F(x + \sqrt{s}G) = \sum_{j=0}^{r} \frac{s^j}{2^j j!} F^{(2j)}(x) + R, \qquad |R| \le \frac{B_{2r+2} s^{r+1}}{2^{r+1}(r+1)!}. \tag{10}$$

Taylor expansion through degree $2r+1$, integration of odd terms, and $\mathbb{E}G^{2r+2} = (2r+2)!/(2^{r+1}(r+1)!)$ prove the formula. It is finite, with a stated derivative hypothesis on the full real line. On panels, derivative control outside the panel is replaced by an explicit tail integral, not discarded. Positivity of the original average does not imply positivity of a truncated differential polynomial. A logarithm of a possibly negative truncation is not an approximation certificate.

If a positive enclosure $0 < L \le \mathbb{E}F \le H$ has been obtained, then $d^{-1}\log L \le W \le d^{-1}\log H$ for $d > 0$. This representation is useful away from $d = 0$, but the apparent $1/d$ sensitivity motivates the amplitude representation below. Polynomial exactness of a forward moment identity is not closure of the nonlinear recursion in a finite polynomial space.

3.3 Source path variation and the entropy support

Along the tilted chain $Z_{-1} = 0$, $Z_i \sim K_i(Z_{i-1}, \cdot)$, the quantities $U_i'(Z_i)$ form a martingale. Consequently their squared means increase with i. Moving a path level lengthens one variance interval and shortens the next. The U'' terms in (9) cancel; the hierarchy coefficients do not:

$$D_q V_f(q)[\dot{q}] = -\tfrac{1}{2}\sum_i w_i \dot{q}_i \, \mathbb{E}U_i'(Z_i)^2, \tag{11}$$

$$D_h \ell(h)[\dot{h}] = -\tfrac{1}{2}\sum_i w_i \dot{h}_i \, \rho_{h,i}, \qquad \rho_{h,i} = \mathbb{E}U_i'(Z_i)^2 \in [0,1]. \tag{12}$$

The terminal field switch uses $(\log\cosh)'' = 1 - \tanh^2$ and the derivative of $-h_{k-1}/2$. Omitting that subtraction loses the Ising entropy normalization. Common refinement, dominated continuity of tilted derivatives, and one-sided feasible segments extend these formulas to repeated levels. For smooth f they yield

$$|V_f(q) - V_f(p)| \le L_f \left\|q - p\right\|_1, \quad L_f = \tfrac{1}{2}\left\|f'\right\|_\infty^2, \qquad |\ell(h) - \ell(h')| \le \tfrac{1}{2}\left\|h - h'\right\|_1. \tag{13}$$

For bounded globally Lipschitz f, first prove the pattern estimate for smooth mollifications with bounded higher derivatives. Their Lipschitz constants do not increase and they converge uniformly, so supremum-norm nonexpansivity passes the estimate to f. Finite step paths are dense in $\mathcal{Q}$ for L^1, giving the unique continuous extension.

The source proves concavity of ℓ in nondecreasing field *quantiles*. Its argument uses folded Gaussian kernels, the opposite monotonicities in the covariance scores, and a symmetric negative Hessian with nonpositive off-diagonal entries and nonnegative row sums. Such a matrix has quadratic form $\sum_i(\sum_j A_{ij})v_i^2 + \sum_{i<j}(-A_{ij})(v_i - v_j)^2 \geq 0$. This is not concavity in an arbitrary affine mixture of overlap laws. The resulting supporting inequality is

$$\ell(h') \leq \ell(h) - \tfrac{1}{2}\int (h' - h)\rho_h, \qquad S_{\mathrm{I}}(\rho_h) = \ell(h) + \tfrac{1}{2}\int h\rho_h. \tag{14}$$

The second equality follows by adding $\frac{1}{2}\int h'\rho_h$ and taking the supremum over *all* field trials, including common refinements; equality is attained at h. It will be essential in obtaining a valid variational upper bound.

4 Amplitude Moments Without a Zero-Coefficient Singularity

Theorem 4.1 (A bounded-state moment enclosure)**.** *Let U be any Borel function taking values in $[a, b]$, set $c = (a + b)/2$, $K = (b - a)/2$, and use any probability measure P for the argument of U. Put $\mu_j = P[(U - c)^j]$ and*

$$\psi_d(t) = \begin{cases} (e^{dt} - 1)/d, & d > 0, \\ t, & d = 0. \end{cases}$$

Suppose $p \geq 1$ and $|\widehat{\mu}_j - \mu_j| \leq \eta_j$ for $1 \leq j \leq p$. Define

$$A_p = \sum_{j=1}^{p} \frac{d^{j-1}}{j!}\widehat{\mu}_j, \qquad I_d = [\psi_d(-K), \psi_d(K)], \qquad \bar{A}_p = \Pi_{I_d} A_p.$$

Here the coefficient for $j = 1$ equals one also at $d = 0$. With $L_d(z) = \log(1 + dz)/d$ for $d > 0$ and $L_0(z) = z$,

$$\left| \frac{\log P(e^{dU})}{d} - (c + L_d(\bar{A}_p)) \right| \leq e^{dK} \left\{ \sum_{j=1}^{p} \frac{d^{j-1}\eta_j}{j!} + \frac{d^p e^{dK} K^{p+1}}{(p+1)!} \right\}, \tag{15}$$

At $d = 0$, interpret the quotient $d^{-1}\log P(e^{dU})$ as PU; the left side remains the absolute approximation error. The estimate holds uniformly for $0 \leq d \leq 1$ with the displayed continuous endpoint interpretation.

Proof. The finite Taylor remainder for e^{dt} gives

$$\left| \psi_d(t) - \sum_{j=1}^{p} d^{j-1}t^j/j! \right| \leq d^p e^{dK}|t|^{p+1}/(p+1)!, \qquad |t| \leq K.$$

Integrate and add the moment errors. The exact value $A = P\psi_d(U - c)$ belongs to I_d, so metric projection on that interval cannot increase its distance from A. On I_d, $L_d'(z) = (1 + dz)^{-1} \leq e^{dK}$; at $d = 0$ the derivative is one. Finally $Pe^{dU} = e^{dc}(1 + dA)$ proves the claim. In particular the log argument after projection is at least $e^{-dK} > 0$. □

Apply this theorem with $P = \mathcal{N}(x, s)$ at each pattern step. The moment coordinate is the *pushforward amplitude* $U - c$, which has compact support even when its Gaussian argument has infinite support. No derivative or analyticity of U is needed for this amplitude Taylor remainder.

This differs from the spatial Taylor formula (10); estimating the moments $\mu_j(x)$ is still a genuine integration task, especially for arbitrary Borel functions. There is no universal finite sample algorithm for all such functions.

The choice of c makes the estimate invariant under constant shifts of U. In implementation, `expm1` and `log1p` or rigorous equivalents avoid subtraction at small d, and rounding errors must be included in the interval budget. Projection guarantees admissible range, not high accuracy by itself.

Proposition 4.2 (Propagation through an implemented hierarchy)**.** *For a fixed finite overlap path, suppose each numerical stage is range-projected into $[a, b]$ and differs from the exact transform applied to its implemented input by at most ϵ_i uniformly over the required real argument. If the implemented terminal has supremum error δ, the final recursion error is at most $\delta + \sum_i \epsilon_i$.*

Proof. Monotonicity and constant-shift equivariance imply $\|\mathcal{T}U - \mathcal{T}V\|_\infty \le \|U - V\|_\infty$: compare $V - \delta \le U \le V + \delta$. The exact transforms preserve $[a, b]$, so projection does not enlarge an error from an exact intermediate. Induct on the finite stages and use the triangle inequality. □

There is no multiplicative instability factor per hierarchy level, although the sum may grow with the number of levels. Uniform infinite-hierarchy accuracy requires a summable allocation of stage errors, not merely a fixed tolerance at each stage. The theorem's uniform real-argument requirement cannot be replaced by checking only a plotting grid.

4.1 Tail costs and the unbounded entropy terminal

For bounded U, truncating a standard Gaussian integration variable to $|G| \le R$ and replacing the removed values by any value in $[a, b]$ gives an error at most $(b-a)P(|G| > R)$ in μ_1; the analogous jth amplitude-moment error is at most $2K^j P(|G| > R)$. These explicit moment errors can enter (15). Conditioning on the retained interval changes the probability measure and must also be budgeted.

The field terminal $\log\cosh x - H/2$ is unbounded, so Theorem 4.1 is *not* applicable to it with an invented finite amplitude range. Each finite field recursion has at most linear growth and Gaussian exponential tails. For example if $U(y) \le B + |y|$, then for $d > 0$ and $|x| \le X$,

$$\mathbb{E}[e^{dU(x+\sqrt{s}G)}; |G| > R] \le 2e^{d(B+X)+d^2 s/2}\overline{\Phi}(R - d\sqrt{s}). \tag{16}$$

Split the Gaussian into its two half-lines and complete the square to obtain the bound. At $d = 0$, expectation tails use $\mathbb{E}[|G|; |G| > R] = 2\varphi(R)$ and the relevant linear-growth bound. Intermediate growth bounds can be propagated through the finite transforms, for instance using $\mathbb{E}e^{d\sqrt{s}|G|} = 2e^{d^2 s/2}\Phi(d\sqrt{s})$. These facts support positive integral enclosures on specified argument ranges. They do not automatically provide all the uniform enclosures required by Proposition 4.2.

5 A Finite-Trial Enclosure of the Infinite Variational Value

The following result concerns the deterministic functional $\mathcal{P}_f$ in (5). Identifying it with an infinite-volume pressure remains conditional on the source theorem. Suppose f is smooth bounded with bounded first derivative, so (13) holds. Finite step paths are enough for a dense overlap grid in L^1.

Theorem 5.1 (Certified lower bound and supporting-field upper bound)**.** *Let $\mathcal{G} \subset \mathcal{Q}$ be a finite δ_q-net for $\mathcal{Q}$ in L^1. Let $\mathcal{H}$ be a nonempty finite family of field steps satisfying $0 \le h \le H$. Suppose*

the evaluated values have uniform errors $|\widehat{V}(q)-V_f(q)| \le \epsilon_V$ *and* $|\widehat{\ell}(h)-\ell(h)| \le \epsilon_\ell$ *on the respective finite families. Define*

$$D = \min_{q\in\mathcal{G}} \max_{h\in\mathcal{H}} \left\{ \alpha\widehat{V}(q) + \widehat{\ell}(h) + \tfrac{1}{2}\int hq \right\}, \quad E = \alpha\epsilon_V + \epsilon_\ell, \quad L = \alpha L_f + H/2.$$

Then $D - E - L\delta_q \le \mathcal{P}_f(\alpha)$.

Independently, choose any admissible field step h_* *bounded by* H_* *and its exact path* ρ_{h_*} *from* (12). *Suppose* $\widehat{q} \in \mathcal{Q}$ *satisfies* $\|\widehat{q} - \rho_{h_*}\|_1 \le \delta_\rho$, *and evaluate* $V_f(\widehat{q})$ *and* $\ell(h_*)$ *with errors* $\epsilon_V^*, \epsilon_\ell^*$. *Then*

$$\mathcal{P}_f(\alpha) \le \alpha\widehat{V}(\widehat{q}) + \widehat{\ell}(h_*) + \tfrac{1}{2}\int h_*\widehat{q} + (\alpha L_f + H_*/2)\delta_\rho + \alpha\epsilon_V^* + \epsilon_\ell^*. \tag{17}$$

Proof. Write $F(q,h) = \alpha V_f(q) + \ell(h) + \frac{1}{2}\int hq$ and $A(q) = \max_{h\in\mathcal{H}} F(q,h)$. Since this is a restriction of the entropy supremum, $\alpha V_f(q) + S_{\mathrm{I}}(q) \ge A(q)$ for all q. Every $F(\cdot,h)$ is L-Lipschitz in L^1, and the same is true of the finite maximum A. For every q choose $q_g \in \mathcal{G}$ with distance at most δ_q. Then $A(q) \ge A(q_g) - L\delta_q \ge D - E - L\delta_q$. Taking the infimum proves the lower bound. No upper bound on all entropy optimizers has been assumed.

For the upper bound, test the outer infimum at the exact $q_* = \rho_{h_*}$. Equation (14) gives the *full* entropy there, not a truncated maximum. Replace q_* by $\widehat{q}$ only in V_f and in the linear pairing; (13) and $\|h_*\|_\infty \le H_*$ bound their changes. Add the two evaluation errors. □

A concrete finite net is obtained by averaging a monotone path on m equal intervals and rounding its values down to multiples of $1/n$. The result stays nondecreasing and $[0,1]$-valued. The averaging error is at most $1/m$: on each interval it is bounded by the interval length times that path's oscillation, and the sum of these oscillations is at most one. Rounding adds at most $1/n$. Thus one may take $\delta_q = 1/m + 1/n$. The net is finite but can be exponentially large. The theorem is a verification specification, not a polynomial-time algorithm.

The lower bound needs no assertion that $\mathcal{H}$ is dense: even a small set gives a valid, possibly weak bound. Enlarging the family improves the exact lower construction. The upper bound is more demanding because ρ_{h_*} must be certified. Its finite-dimensional components are expectations of bounded $U_i'^2$, but their tilted-chain normalizers must also be controlled. Bounds on those components and monotonicity can be combined with interval isotonic constraints. A stationary point of a finite numerical minimization alone is neither the global infimum nor an entropy attainment certificate.

For $|f|, |g| \le K$, the source's uniform Gaussian-L^1 bound extends the enclosure from a smooth g to a bounded Borel f:

$$|\mathcal{P}_f(\alpha) - \mathcal{P}_g(\alpha)| \le \alpha e^{2K} \|f - g\|_{L^1(\gamma)}. \tag{18}$$

One must obtain this approximation error by a separate argument. It is not a derivative bound for f. The state range can be centered to improve constants, but cannot be assumed finite for a hard constraint whose log weight is $-\infty$.

6 Disorder Replacement Beyond Symmetric Signs

Theorem 6.1 (Explicit finite-size replacement)**.** *Let* ξ *be real with* $\mathbb{E}\xi = 0$, $\mathbb{E}\xi^2 = 1$ *and* $m_3 = \mathbb{E}|\xi|^3 < \infty$. *Give each pattern entry an independent copy of* ξ. *If* $f \in C^3(\mathbb{R})$ *and* $B_j = \left\|f^{(j)}\right\|_\infty < \infty$ *for* $j = 1,2,3$, *with finite expected finite-volume pressures, then*

$$|\mathbb{E}p_N^\xi(f) - \mathbb{E}p_N^G(f)| \le \frac{M_N}{6N^{3/2}}(B_3 + 6B_1B_2 + 8B_1^3)\left(m_3 + 2\sqrt{2/\pi}\right). \tag{19}$$

In particular the right side is $O(N^{-1/2})$ at fixed density.

Proof. Hold every entry except one fixed and write its value as t. Only one row contribution changes. Three derivatives of the finite log sum give

$$\partial_t^3 p_N = N^{-5/2}\left\{\langle x_i^3 f'''\rangle + 3\,\mathrm{Cov}(x_i^2 f'', x_i f') + \left\langle (x_i f' - \langle x_i f'\rangle)^3 \right\rangle\right\}.$$

Here all derivatives of f are at that row's field and all averages are under the current Gibbs measure. Since $|x_i| = 1$, the absolute value is at most $N^{-5/2}(B_3 + 6B_1B_2 + 8B_1^3)$. Taylor expansion at zero through degree two has remainder at most this bound times $|t|^3/6$. The expectations of the first two Taylor terms match for ξ and G. Replace all $M_N N$ entries, always conditionally on the others, and sum the errors. Finally $\mathbb{E}|G|^3 = 2\sqrt{2/\pi}$. □

Corollary 6.2 (Bounded almost-everywhere-continuous disorder extension)**.** *Under the disorder assumptions of Theorem 6.1, let f be bounded Borel with a Gaussian-null discontinuity set. If the source Gaussian pressure theorem holds, then both $\mathbb{E}p_N^\xi(f)$ and $p_N^\xi(f)$ in probability converge to $\mathcal{P}_f(\alpha)$.*

Proof. For every deterministic cube state x, the fresh field is $X_{N,x} = N^{-1/2}\sum_i x_i\xi_i$. Uniformly over x, the Lyapunov sum of third absolute moments is $m_3/\sqrt{N}$, while the variance is one and the mean is zero. The Lindeberg–Feller central limit theorem therefore implies convergence to a standard Gaussian along *every* sequence of deterministic sign choices. Equivalently, the convergence of bounded Gaussian-almost-everywhere-continuous tests is uniform over those choices: otherwise a violating subsequence and its selected sign vectors would contradict that theorem. No symmetry of ξ is used.

Choose $g_j \in C_c^\infty$ with $|g_j|, |f| \le K$ and $\|g_j - f\|_{L^1(\gamma)} \to 0$. Remove one whole pattern before differentiating the interpolation between f and g_j. Restoring its bounded log factor changes a nonnegative average by at most e^{2K}. Hence

$$|\mathbb{E}p_N^\xi(f) - \mathbb{E}p_N^\xi(g_j)| \le \frac{M_N}{N} e^{2K} \sup_x \mathbb{E}|f(X_{N,x}) - g_j(X_{N,x})|.$$

For each fixed j, the supremum tends to $\|f - g_j\|_{L^1(\gamma)}$ by the uniform test convergence. Apply Theorem 6.1 to g_j, then the Gaussian theorem and (18), first letting $N \to \infty$ and then $j \to \infty$.

For probability convergence, changing one pattern row changes p_N by at most $\mathrm{osc}(f)/N$, independently of entry tails. The bounded-differences inequality gives

$$\mathbb{P}(|p_N^\xi - \mathbb{E}p_N^\xi| \ge t) \le 2\exp\left(-\frac{2N^2t^2}{M_N\,\mathrm{osc}(f)^2}\right)$$

when $M_N > 0$ and $\mathrm{osc}(f) > 0$. Constant activations and zero patterns are deterministic. This proves the final assertion. □

This is an extension of the *stated random-sign conclusion* in the assigned manuscript using the classical replacement principle [9], not a claim that third-moment universality is new in spin-glass or perceptron theory. No explicit rate is claimed for arbitrary bounded almost-everywhere-continuous f: approximation moduli and growth of derivatives of g_j would be needed. Independent but nonidentical triangular arrays, fewer moments, correlated patterns, and hard constraints need additional uniform integrability or dependence arguments.

The discontinuity condition is substantive. For $f = \mathbf{1}_{\mathbb{Q}}$, Gaussian fields have $f = 0$ almost surely at all finitely many spin states. With symmetric sign entries and square N, all fields are

rational and the pressure is M_N/N; with odd nonsquare N all numerator sums are nonzero odd integers and the fields are irrational. There is no common limit. Moment coordinates do not remove this obstruction.

7 Source Proof Architecture and the Untransferred Tasks

The recursions are only the analytic part of the source's claim. Its other sections address a highly nontrivial probabilistic identification. The spin overlap $R_{ab} = N^{-1}x^a \cdot x^b$ must be generated by the actual nonlinear model; it cannot simply be chosen to optimize a Gaussian formula.

The array step uses perturbations, Ghirlanda–Guerra identities, duplication and ultrametricity to obtain marked cascade limits. Visits to the same cascade leaf still receive independent residual Gaussian noise. Sharing that residual between replicas changes the covariance and the restored-pattern calculation. A finite covariance matrix test cannot substitute for the array convergence and conditioning assertions.

The upper bound enriches configurations with cascade labels and hierarchical Gaussian fields and Poissonizes pattern counts. At a deterministic negative contact minimum, the quadratic perturbation penalty must enforce fluctuation control and the joint identities. Synchronization orders the spin and label overlaps, and field variations yield tail inequalities for the density derivative. A small numerical gradient at a trial point is not that deterministic contact-point construction.

The bulk step deletes two patterns before restoring them. Both first and second moments are needed to identify the empirical gradient covariance as a deterministic function of the overlap and to control its diagonal correction. A first-moment representation alone does not give the conditional concentration used by the cavity argument.

The cavity step first adds a fixed number L of coordinates, controls Taylor and normalized-measure errors as $N \to \infty$, and identifies the cavity overlap with the bulk one by coordinate exchangeability. Only then is L allowed to grow in the telescoping lower-bound argument. Entropy attainment uses (14) at that actual path; it is not attainment at an arbitrary local optimizer.

For general bounded activations, deleting a fresh pattern gives a dimension-uniform Gaussian-L^1 approximation bound. For the recursion, the product tilt telescopes to

$$\log \frac{dP_{\text{tilted}}}{dP_{\text{Gaussian}}} = f_t(Z_k) - \sum_{i=0}^{k-1} w_i U_i(Z_i),$$

bounded by $2K$ when all functions lie in $[-K, K]$. The unweighted endpoint is standard normal because the total variance is one. Thus the approximation is uniform in the hierarchy, despite varying numbers of levels. This supplies (18); a formal derivative series for a Borel f would not.

For the source's unbounded extension, the assumptions are $\mathbb{E}|f(G)| + \mathbb{E}e^{f(G)} < \infty$. With $f_{J,K} = \min(K, \max(-J, f))$, the displayed error is

$$\varepsilon_{J,K} = \mathbb{E}(-J - f(G))_+ + e^J \mathbb{E}(e^{f(G)} - e^K)_+.$$

The order is $K \to \infty$ at fixed J, then $J \to \infty$, after the bounded finite-volume limit. Reversing this order can destroy the upper-tail bound. Neither this statement nor our bounded-state amplitude theorem includes log-zero constraints without further analysis.

8 Checks, Examples, and Reproducibility

For $f = c$ constant the exact pressure is $M_N c/N$ and the variational value is αc. For the integrable linear activation $f(s) = ts$, the recursion gives

$$V_f(q) = \frac{t^2}{2}\left(1 - \int_0^1 q\right).$$

Let $H = \alpha t^2$ and $q_* = \mathbb{E}\tanh^2(\sqrt{H}G)$. The field support equality at the constant field H gives $\mathcal{P}_f(\alpha) = \mathbb{E}\log\cosh(\sqrt{H}G)$. The direct finite model also factors into independent Gaussian coordinate-field cosh terms with variance $t^2 M_N/N$. These checks detect missing $\log 2$, endpoint variance, and diagonal-shift terms; they are not proofs for general nonlinear activations.

An independent source reviewer read all 3,652 lines of the 11 assigned TeX/BibTeX files and matched their bytes to the pinned source records. Its bounded contract review found no blocking mathematical error. In particular, it checked the deterministic contact-point fluctuations, conditional marks and per-visit residuals, distinct-row second moments, the normalized cavity comparison, entropy measurability, and the stated limit order. This is not a formal proof certificate or a reproduction of every external dependency. The parent directly read the model, introduction, variational and extension sections, bibliography, and selected bulk passages; the broader source review is explicitly attributed to that independent reviewer.

An independent local-result review checked the amplitude, chain, bracket, replacement, and disorder-extension proofs. It found no blocking error in their intended statements; its endpoint wording correction has been incorporated. The reproducible accompanying script passed nine diagnostic groups. Four groups contain 9,704 exact symbolic or rational cases, including projection, net coverage, third-derivative identities, and a scalar supporting-entropy model. Four further groups contain 853 high-precision, non-interval cases for amplitude errors, finite chains, derivatives, and one-pattern linear-activation replacement. One mixed group checks 258 signed-binomial distributions for a genuinely asymmetric two-point disorder law, with exact integer probabilities and non-interval Gaussian comparisons. The large count is dominated by small-grid enumeration and is not a count of independent theorem proofs.

These diagnostics do not certify Gaussian quadrature errors, optimized Ising variational intervals, all hierarchy paths, or infinite-volume limits. The scalar toy entropy is not the Ising entropy. The general claims rest on the displayed analytic proofs and their stated source dependencies, not on finite numerical agreement. No empirical runtime comparison or certified optimized free-energy interval for a general activation is asserted here.

9 Current Literature and Version Boundaries

The primary PMLR publication of Bolthausen, Nakajima, Sun, and Xu [5] establishes a small-density replica-symmetric formula through moment methods conditioned on approximate message passing, under its activation conditions. The assigned manuscript's all-density hierarchical assertion is substantially broader and needs its own proof. The recent centered-Ising path-convexity work of Chen, Issa, and Mourrat has version 2 dated September 29, 2026 [6]; Ho's June 2026 Parisi-PDE concavity preprint [7] supplies another route to the relevant analytic structure. These are source-related inputs, not a direct proof of nonlinear-pattern pressure from Gaussian mixed-spin covariance.

Shmalo's July 2026 preprint [8] claims an Arb-assisted verification of the numerical conditions in zero-margin Ising storage-capacity results. Its claim and scope are distinct from positive-

temperature bounded real log-potentials. The programs have not been reproduced in this report, and a metadata check is not an independent proof audit.

The public upstream history was checked on October 8, 2026. A complete tree comparison of the pinned edition with commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb` found no changed manuscript in the selected 100-report queue. Other papers were repaired or withdrawn. Thus the fixed source edition is retained with explicit provenance, not treated as proof that all statements in the collection are established.

10 Advantages, Disadvantages, and Further Research

Step	Source method	Ram reformulation and its boundary
Gaussian transform	Normalized exponential integral and heat generator.	Complete derivatives retain covariance terms; amplitude moments give an explicit zero-coefficient-safe remainder. Moment integration remains necessary.
Hierarchy	Exact finite recursions and continuity in overlap paths.	One-step errors add under nonexpansiveness. This does not make a fixed error tolerance adequate at infinitely many levels.
Entropy dual	Concavity and a supporting field identify attainment at a generated overlap path.	Finite lower trials and a certified supporting-field upper trial enclose the same functional. No unjustified finite entropy cap is used for an upper bound.
Disorder	Entry replacement for smooth activations; approximation for symmetric signs.	Explicit constants and a finite-third-moment i.i.d. extension with Gaussian-null discontinuities. No claim for all Borel data, correlated disorder, or hard constraints.
Thermodynamic proof	Contact minima, identities, two-pattern restoration and cavity limits.	The added certificates can help check analytic substeps, but do not prove these probabilistic identifications.

The advantages are auditable error budgets, explicit domains and normalizations, stable treatment of small hierarchy coefficients, and a clear distinction between a valid variational bound and a numerical optimizer. The disadvantages include large finite nets, high-order moment conditioning, field-tail costs, and the need for certified derivatives or integrals rather than sampled plots. The lower enclosure may remain weak when the chosen field family misses important dual directions. The upper enclosure may require difficult tilted-chain overlap integrals. There is no measured speed or accuracy dominance over established quadrature, Parisi-PDE, interpolation, or approximate-message-passing methods.

A useful next step is an interval implementation for a specified smooth activation, with positive Gaussian quadrature, independent tail bounds, and lower/upper certificates from Theorem 5.1. Adaptive monotone-path subdivision could reduce the finite-net cost, but would need a proof that excluded cells cannot improve the lower bound. Another target is quantitative smoothing for bounded activations with a known modulus or bounded variation, yielding an explicit combination of approximation and disorder-replacement rates. Reaching hard constraints or zero temperature

would require uniform control in the diverging activation range, precisely where the present amplitude constants and clipping estimates deteriorate.

11 Conclusion

Ram Calculus supplies a concrete moment-and-remainder representation for the source's normalized Gaussian recursions and a useful way to organize complete derivatives, uncertainty propagation, and variational certificates. The independent additions are the bounded-amplitude enclosure, its finite-hierarchy propagation, the lower/supporting-field upper bracket, and the explicit disorder-replacement extension under stated assumptions. Their contribution is assessed as C3/5 while applicability remains L4/5. These are neither a declaration of a new unconditional solution of the entire perceptron problem nor an assertion of historical novelty for the classical mathematical tools.

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